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ENVENOMATION INJURIES

ENVENOMATION CAUSES LOCALISED (OEDEMA/MYOTOXIC) and/or SYTEMIC TOXICITY (NEUROTOXIC, HAEMOTOXIC OR CARDIOTOXIC). DEFINITIVE TREATMENT IS ANTIVENOM. CONSIDER ANAPHYLAXIS IN ACUTE ABC CONCERNS. EARLY EVAC AND ANTIVENOM IS MAINSTAY OF TREATMENT.

PRIMARY SURVEY

AIRWAY	ACUTE AIRWAY OEDEMA AND STRIDOR ARE TREATED AS ANAPHYLAXIS
BREATHING	NEUROTOXICITY CAN CAUSE RESPIRATORY PARALYSIS SO VENTILATE PATIENT WITH RATE <10
CIRCULATION	CARDIOTOXIC CAN CAUSE ARRYTHMIA SO EARLY CONTINUOUS ECG MONITORING AND PREPARE FOR ALS

THINK ANAPHYLAXIS IN PATIENTS WITH ACUTE SYMPTOMS OF AIRWAY SWELLING, DYSPNOEA AND SHOCK

ASSESSMENT – IDENTIFY SOURCE, TIME AND LOCATION OF BITE/STING

FULL HEAD TO TOE WITH VITAL SIGNS AND ECG – REPEAT EVERY 15 MINUTES FOR SYMPTOM PROGRESSION
FULL CRANIAL NERVE AND MOTOR FUNCTION ASSESSMENT FOR SIGNS OF NEUROTOXICITY
20 MINUTE WHOLE BLOOD CLOTTING TEST (20WBCT)* FOR HAEMOTOXICITY (BELOW)
ASSESS SIZE OF OEDEMA OR NECROSIS AND MARK WITH SKIN PEN

LOCALISED SIGNS	SYSTEMIC SIGNS		
OEDEMA/MYOTOXIC	NEUROTOXIC	HAEMOTOXIC	CARDIOTOXIC
ERYTHEMA ECCHYMOSIS BLISTERING INCREASING PAIN SWELLING/OEDEMA NECROTIC TISSUE COMPARTMENT SYNDROME	PTOSIS (EYE DROOP) OPTHALMOPLEGIA DIPLOPIA (BLURRY SIGHT) BULBAR WEAKNESS DYSPNOEA (BREATHING) PROGRESSIVE PARALYSIS RESPIRATORY PARALYSIS	BLEEDING FROM WOUND SPONTANEOUS BLEEDING FROM HEALTHY MUCOSA HAEMATEMESIS SHOCK DIC ACUTE KIDNEY INJURY	PALPITATIONS ARRYTHMIA (LONG QT, BRADYARRYTHMIA, FAST ATRIAL FIBRILLATION) MYOCARDIAL INFARCTION HYPOTENSION CARDIOGENIC SHOCK

REQUEST IMMEDIATE MEDICAL EVACUATION TO NEAREST MTF OR HOSPITAL THAT HAS ANTIVENOM

IMMEDIATE MANAGEMENT

AIRWAY	MANAGE AIRWAY AND VENTILATIONS - BE PREPARED ANAPHYLAXIS
BREATHING	ADMINISTER OXYGEN TO MAINTAIN OXYGEN SATURATIONS >94%
CIRCULATION	ECG MONITORING FOR ARRYTHMIA AND BP FOR CARDIOVASCULAR STATUS GAIN IV/IO ACCESS AND CONSIDER FLUID CHALLENGE IF HYPOTENSIVE ASSESS FOR COMPARTMENT SYNDROME (OEDEMA) AND CONSIDER FASCIOTOMY IMMOBILISE LIMB AND APPLY ECB IN A DISTAL-PROXIMAL-DISTAL METHOD WITH SPLINTING TO AVOID INCREASING BLOOD FLOW THAT COULD WORSEN SYSTEMIC REACTIONS
DISABILITY / HANDLING	CONSIDER ANALGESIA FOR PATIENTS LOOSEN ANY TIGHT FITTING CLOTHING OR ITEMS AROUND BITE DUE TO SWELLING
DRUGS	ADRENALINE 1:1000 500mcg EVERY 5 MINUTES FOR ANAPHYLAXIS ATROPINE 600mcg - BRADYCARDIA AND HYPERSALIVATION (NEUROTOXIC) NEOSTIGMINE 500mcg - NEUROTOXIC SIGNS CEFTRIAXONE or MEROPENEM - OPEN SORES or NECROSIS CONSIDER BLOOD PRODUCTS FOR HAEMOTOXIC EFFECTS OF COAGULOPATHY RB

ANTIVENOM ADMINISTRATION

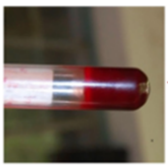
INDICATIONS	SIGNS OF SYSTEMIC TOXICITY (SEE ABOVE), SHOCK (HYPOTENSION), RAPIDLY PROGRESSING OEDEMA OR BLISTERING, OR ANY SIGNS OF RENAL FAILURE (HAEMATURIA).
PROCEDURE	PRE-TREAT WITH PROPHYLACTIC DOSE OF 250mcg 1:1000 ADRENALINE SUBCUTANEOUSLY ADMINISTER INTRAVENOUSLY AS DIRECTED ON INSTRUCTIONS IF ANAPHYLAXIS OCCURS STOP INJECTION AND TREAT WITH 500mcg 1:1000 ADRENALINE IM EVERY 5 MINUTES UNTIL SYMPTOMS IMPROVE. CONSIDER FLUIDS AND ANTIHISTAMINES.

RB

***20WBCT**

COLLECT 2ml OF BLOOD IN A STERILE GLASS TUBE
STAND UPRIGHT IN ROOM TEMP FOR 20 MINUTES
ASSESS TO SEE IF BLOOD HAS CLOTTED

NORMAL



HAEMOTOXIC

POINTS TO NOTE

1. **SNAKEBITES PROVIDE THE GREATEST RISK TO SERVICE PERSONNEL SO THIS GUIDELINE IS MORE FOCUSED ON SNAKE ENVENOMATION BUT CAN BE TRANSFERABLE TO OTHER ENVENOMATIONS.**
2. **THERE ARE USUALLY TWO MAIN TYPES OF VENOMOUS SNAKES, ELAPIDS WHICH MAINLY CAUSE NEUROTOXICITY, AND VIPERS WHICH MAINLY CAUSE HAEMOTOXIC AND CARDIOTOXIC EFFECTS. CARDIOTOXICITY CAN BE CAUSED BY ELAPIDS, HOWEVER, THIS IS RARE.**
3. **ALL BITES ARE TO BE TREATED AS VENOMOUS AND SHOULD BE EVACUATED TO A MTF FOR FURTHER ASSESSMENT AND MANAGEMENT. ANXIETY FROM BITES CAN MIMIC SYMPTOMS SIMILAR TO SYSTEMIC TOXICITY SO REASSURANCE IS SIGNIFICANT IN THE MANAGEMENT OF BITES.**
4. **NEUROTOXIC ENVENOMATION CAN RESULT IN RESPIRATORY PARALYSIS SO PREPARE TO MANAGE THE AIRWAY AND VENTILATE WHILST TRANSFERING TO A MTF WITH ANTIVENOM. THE PATIENT MAY PRESENT ALERT BUT PARALYSED, SO GCS IS NOT A RELIABLE INDICATOR OF SEVERITY.**
5. **COMPARTMENT SYNDROME SHOULD BE SUSPECTED IN OEDEMATOUS AND ERYTHEMATOUS LIMBS THAT HAVE PAIN, PALLOR, PARAESTHESIA, POIKILOTHERMIA, PULSELESSNESS AND PARALYSIS.**
6. **BULBAR WEAKNESS INVOLVES THE CRANIAL NERVES ASSOCIATED WITH THE TONGUE, VOCAL CORDS AND JAW MUSCLES – DYSPHONIA, DYSPHAGIA AND DYSARTHRIA ARE HALLMARK SIGNS.**
7. **BITES TO THE FACE OR NECK MUST BE MONITORED BY A MEDICAL OFFICER.**
8. **TOURNIQUETS ARE NOT TO BE USED IN ENVENOMATION INJURIES.**
9. **ANAPHYLAXIS IS MORE LIKELY TO BE CAUSED BY THE ANTIVENOM, SO STOP INJECTION AND TREAT AS PER ANAPHYLAXIS GUIDELINES OF 500mcg IM ADRENALINE EVERY 5 MINUTES UNTIL SYMPTOMS IMPROVE. 10mg CHLORPHENAMINE AND 200mg HYDROCORTISONE CAN BE CONSIDERED BUT SHOULD NOT DELAY ADMINISTRATION OF ADRENALINE. NEVER PRE-TREAT WITH ANTIHISTAMINE OR STEROIDS.**
10. **PATIENTS THAT HAVE BEEN BITTEN/STUNG AND PRESENT WITH NO SYSTEMIC SYMPTOMS REQUIRE TO BE MONITORED FOR 24 HOURS BEFORE BEING DISCHARGED. ROUTINE FULL MONITORING OF VITALS AND NEUROLOGICAL ASSESSMENT ARE THE MINIMUM STANDARDS.**

Guideline written in accordance with MERT Standard Operating Procedure 'Envenomation V3.0'

References

Ralph, R., et al. (2022) Managing Snakebites. *BMJ*, 376, p. e057926. Doi: https://doi.org/10.1136/bmj-2020-057926

World Health Organisation (WHO). (2016) *Guidelines for the Management of Snakebites,* 2nd Edition. Available at: Guidelines for the management of snakebites, 2nd edition (who.int) (Accessed 13th Feb 2021).

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